THE UNIVERSITY OF TEXAS
MD Anderson Cancer Center
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MD Anderson Dosimetry Lab
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Report of IMRT Head and Neck Phantom Irradiation

Date of Report:	October 29, 2019
Institution:	Cameroon Oncology Center
Physicist:	Paul Mobit / Imandi Paulin
Radiation Machine:	Varian, Clinac 21EX (1813) – 6 MV
Intensity Modulation Device:	Multileaf Collimator
IMRT Technique:	Dynamic MLC
Treatment Planning System:	Varian Eclipse – AAA algorithm
Date of Irradiation:	August 28, 2019

Description of Procedure

An anthropomorphic head phantom incorporating a rectangular dosimetry insert was imaged and irradiated to approximately 6.6 Gy using an IMRT technique. The dosimetry insert consisted of one primary PTV containing four TLD capsules, a secondary PTV and an organ at risk (OAR), each containing two TLD capsules. The TLD capsules provided point dose information. Three sheets of GAFChromic™ Dosimetry Media provided dose profiles through the center of primary PTV.

The dosimetric precision of the TLD is 3%, and the spatial precision of the film and densitometer system is 1 mm.

Summary of TLD and film results:

Location	IROC-H vs. Inst.	Criteria	Acceptable
Primary PTV sup. ant.	0.96	0.93 – 1.07	Yes
Primary PTV inf. ant.	0.96	0.93 – 1.07	Yes
Primary PTV sup. post.	0.96	0.93 – 1.07	Yes
Primary PTV inf. post.	0.96	0.93 – 1.07	Yes
Secondary PTV sup.	0.98	0.93 – 1.07	Yes
Secondary PTV inf.	0.97	0.93 – 1.07	Yes

Film Plane	Gamma Index*	Criteria	Acceptable
Axial	91%	≥85%	Yes
Sagittal	92%	≥85%	Yes

*Percentage of points meeting gamma-index criteria of 7% and 4 mm.

The phantom irradiation results listed in the table above **do meet** the criteria established by IROC Houston in collaboration with the cooperative study groups. Had you been applying for credentialing to enter patients into certain protocols that allow the use of IMRT, your institution **would have** satisfied the phantom irradiation component of the credentialing process.

TLD and Film Analysis by: Andrea Molineu, M.S., and Blake Lewis

Report Checked by:

David S. Followill, Ph.D.
Chief, Section of Outreach Physics

The information provided to you in this report should be considered as a quality assurance peer review and should only be used as a supplement to your institution's own commissioning and quality assurance measurements. Changes should be made by your institution only after you have determined that changes are warranted. Changes should not be made on the basis of this report alone. Such changes, if necessary, should be deliberate, and with the full knowledge of all individuals concerned.

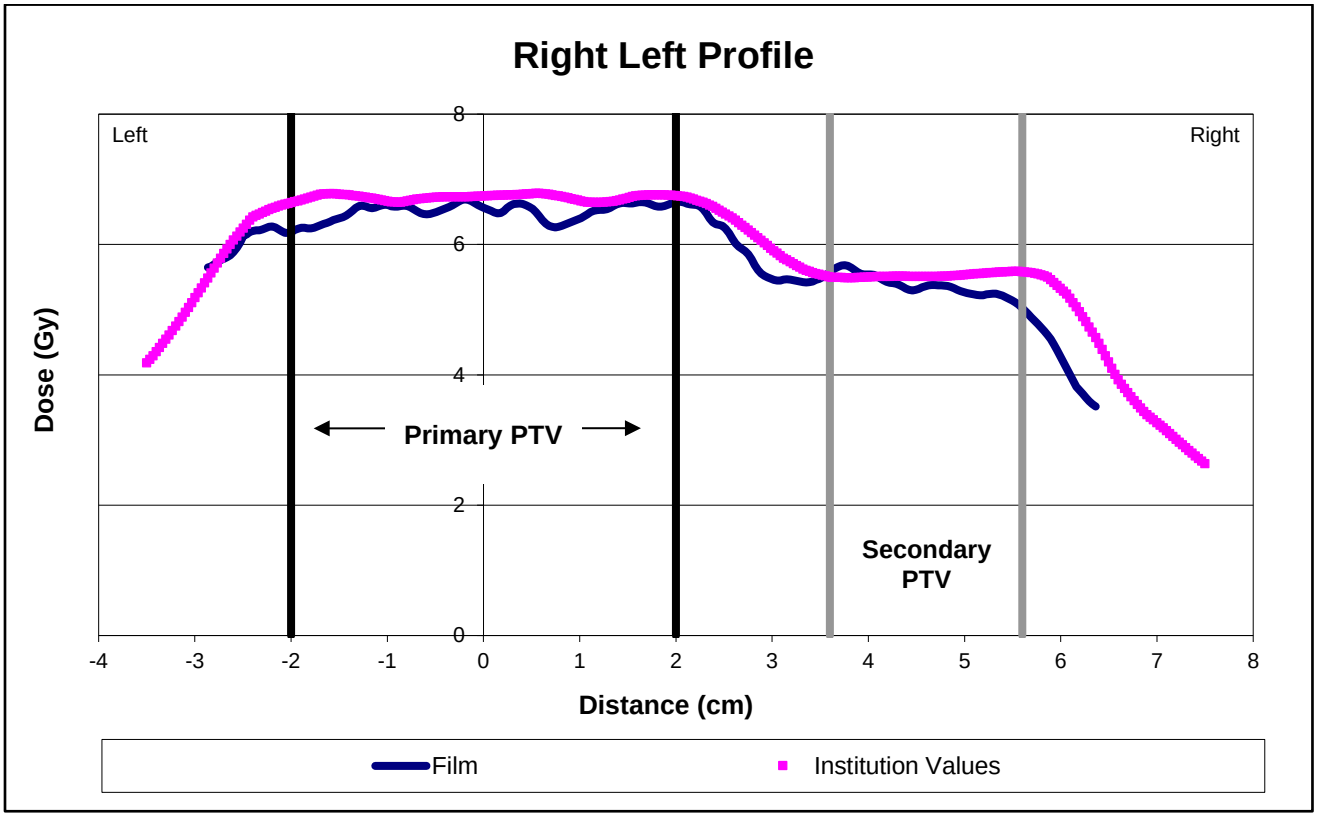
TLD Results:

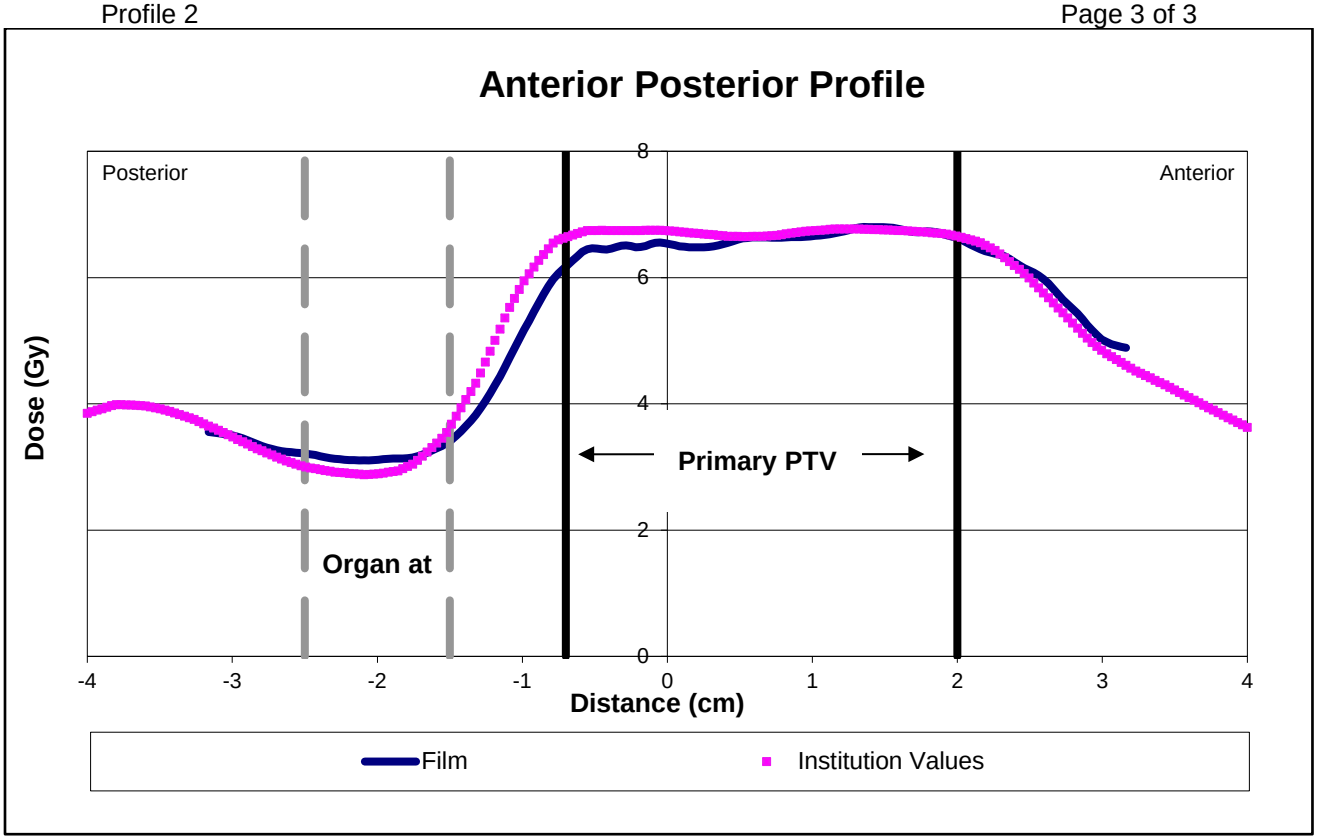
Location	Institution Reported Mean Dose	TLD Dose (cGy)	Measured/Institution
Primary PTV sup. ant.	659	635	0.96
Primary PTV inf. ant.	664	635	0.96
Primary PTV sup. post.	677	652	0.96
Primary PTV inf. post.	674	649	0.96
Secondary PTV sup.	548	535	0.98
Secondary PTV inf.	547	531	0.97
Organ at risk sup.	295	299	1.01
Organ at risk inf.	285	287	1.01

Film results:

Film profiles through the center of the primary PTV were scaled to TLD dose values. The results are plotted with dose information taken from the computer generated isodose lines provided by the institution. The displacement between the measured dose gradient and the institution's calculated dose gradient has been determined in the region between the primary PTV and the OAR. The displacement was measured at three levels and averaged. The upper bound for acceptable agreement is 4 mm.

Profile 1




Anterior Posterior Profile
Posterior
Anterior
Dose (Gy)
Primary PTV
Organ at
Distance (cm)
Film
Institution Values

Profile 3

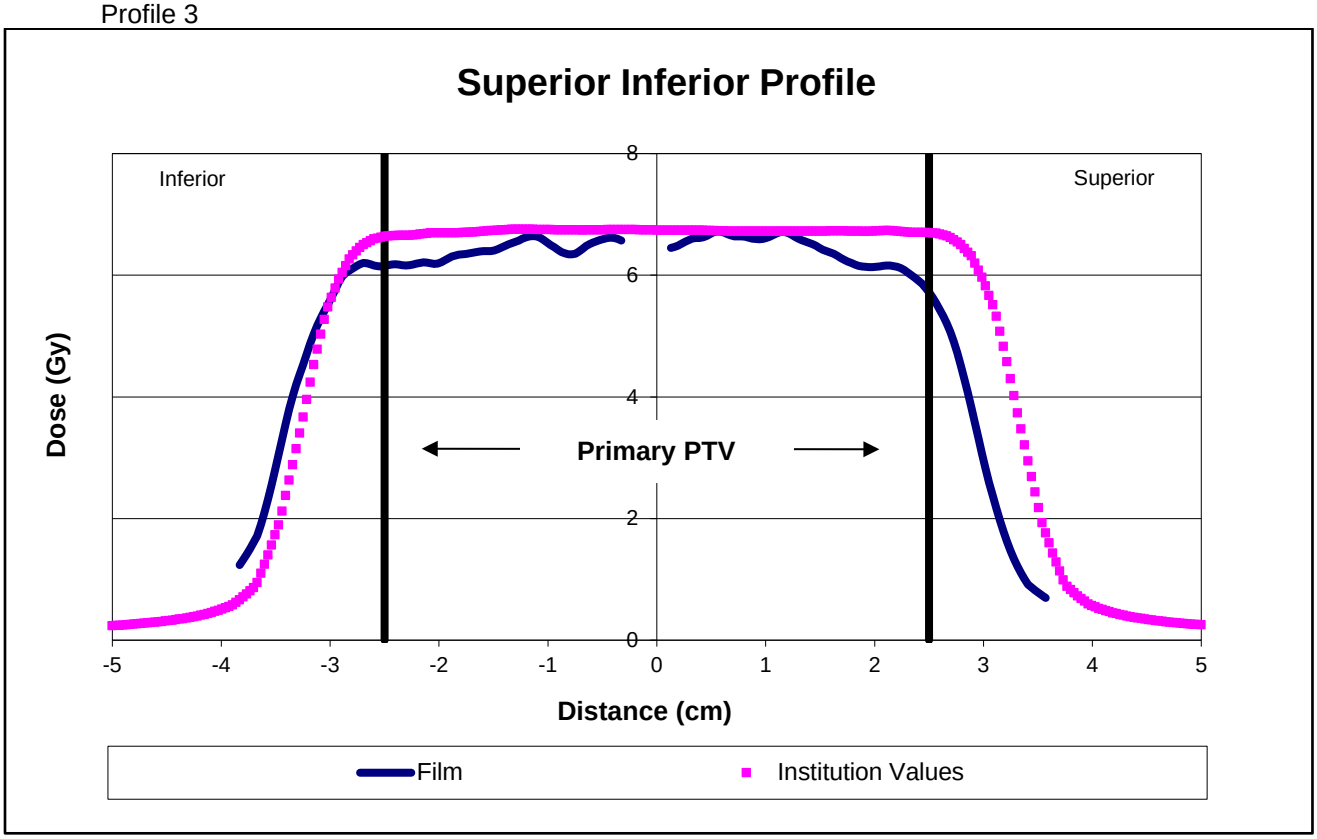

Superior Inferior Profile
Inferior
Superior
Dose (Gy)
Primary PTV
Distance (cm)
Film
Institution Values

Figure 4a: The area on the axial film that was used for gamma analysis. Artifacts have been masked and are not included in the gamma analysis.

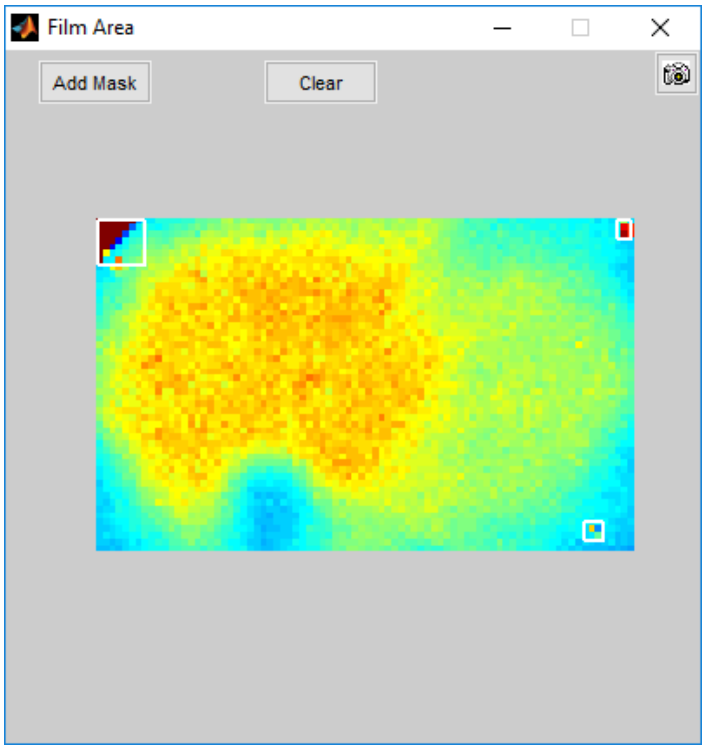
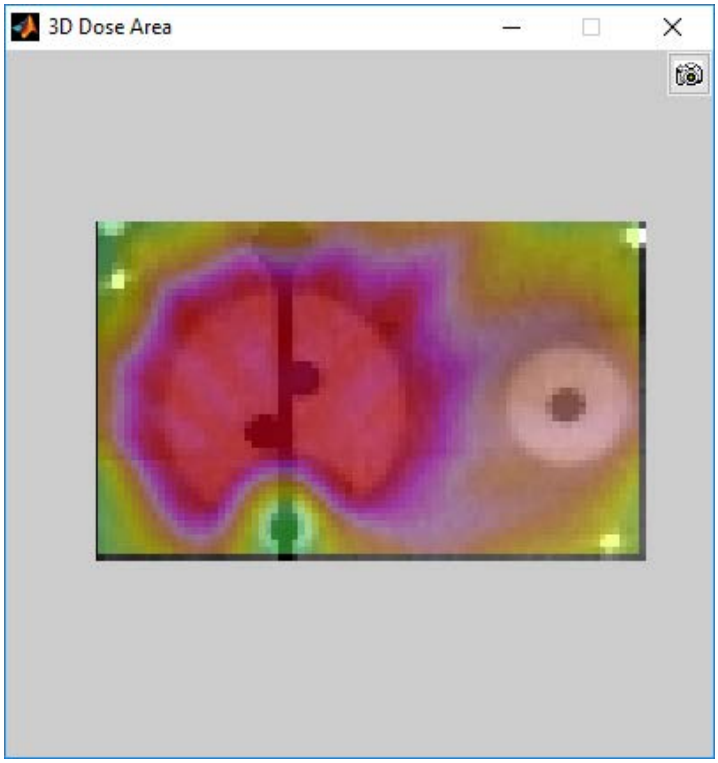


Figure 4b: The corresponding area in the axial plane from the treatment planning system.



Pie charts indicate percentages of pixels passing and failing the specified criteria.

Figure 4c: The result of the axial gamma analysis (color scale).

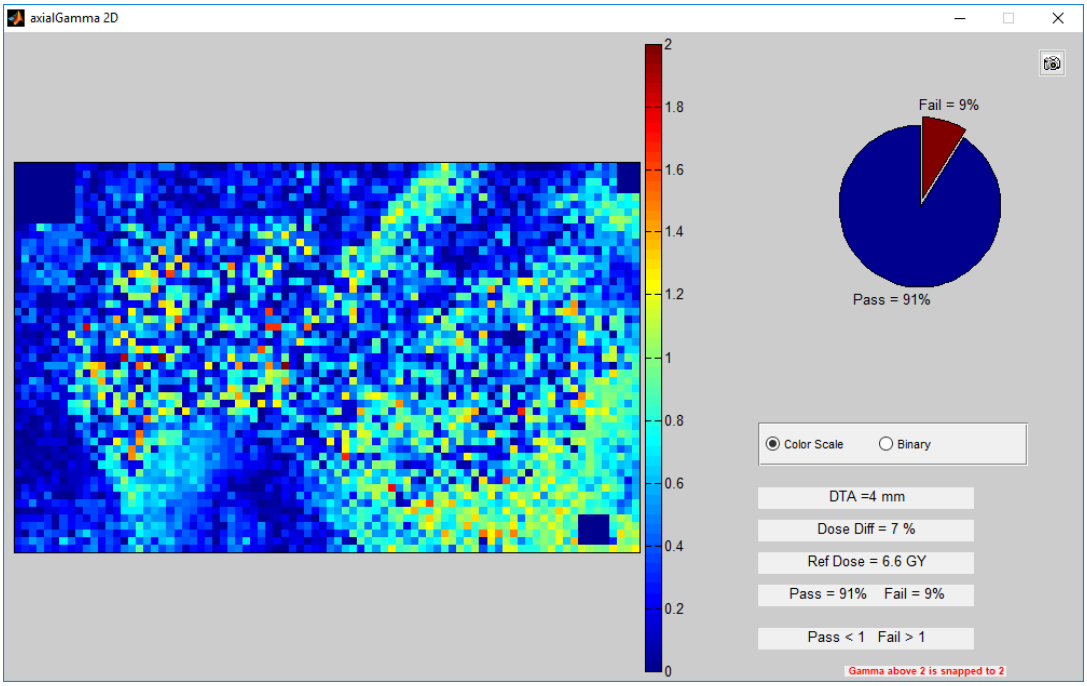


Figure 4d: The result of the axial gamma analysis (binary). The red areas are locations that fail the criteria.

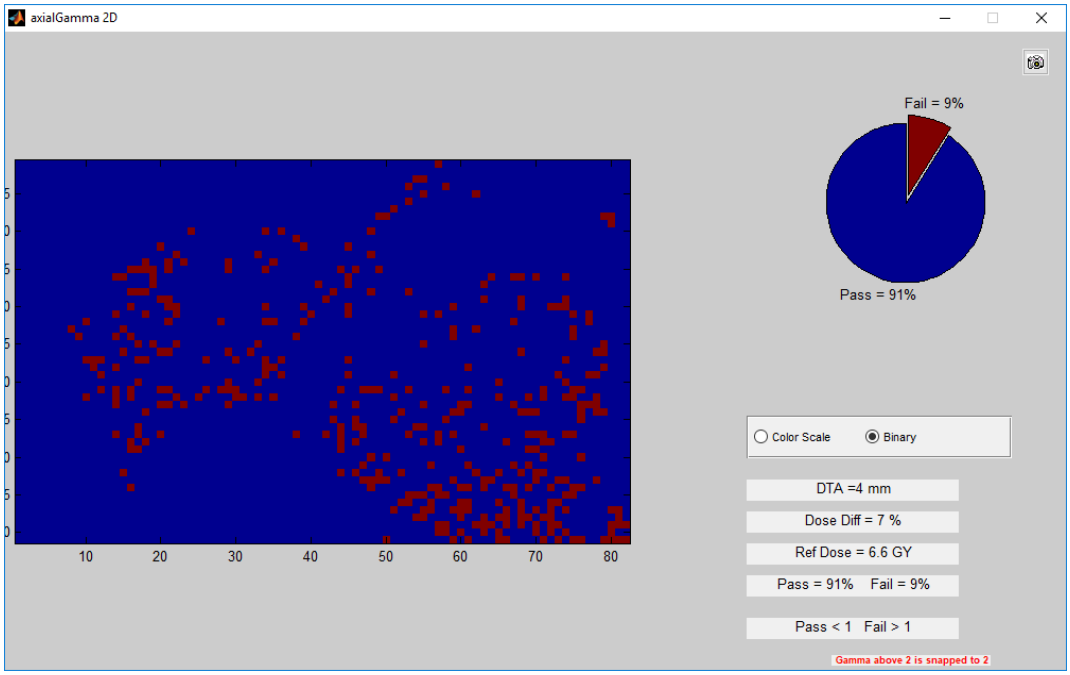


Figure 5a: The area on the sagittal film that was used for gamma analysis. Artifacts have been masked and are not included in the gamma analysis.

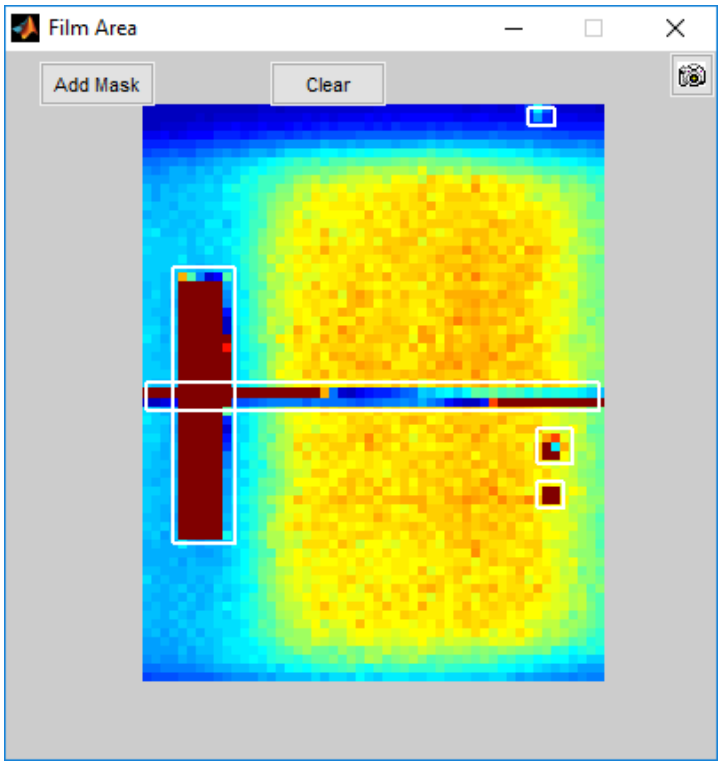
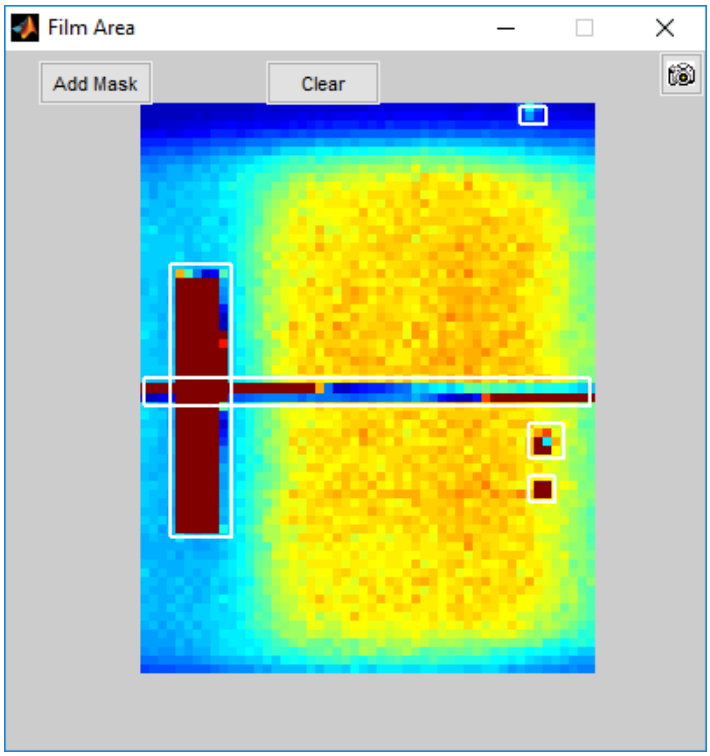


Figure 5b: The corresponding area in the sagittal plane from the treatment planning system.



Pie charts indicate percentages of pixels passing and failing the specified criteria.

Figure 5c: The result of the sagittal gamma analysis (color scale).

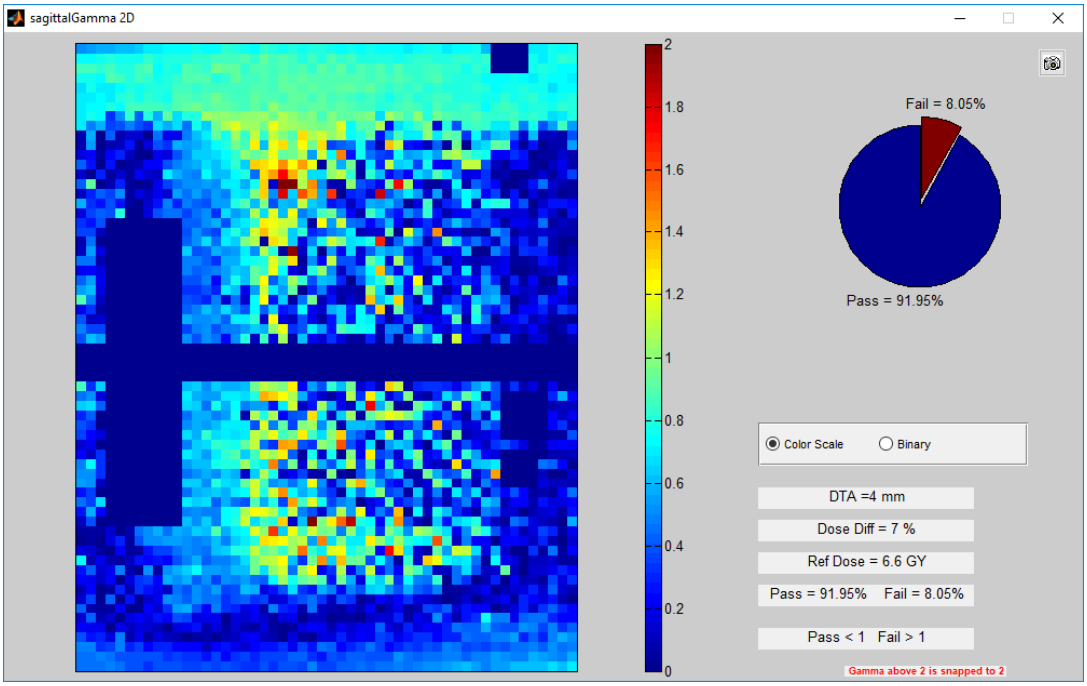


Figure 5d: The result of the sagittal gamma analysis (binary). The red areas are locations that fail the criteria.

